SECTION 14 / 18
VALIDATION

Experiment Design

Three testable hypotheses drawn straight from the teardown — each with a control, a variant, a primary metric, and a guardrail that would tell us to stop.

EXPERIMENTS	DERIVED FROM	EVERY TEST HAS	ROLLS UP TO
3	Sections 9 & 11	A guardrail metric	Weekly Successful AI Tasks

14.1 Experiment 1 — Use-case-first onboarding

Does routing by job beat browsing by product?

ACTIVATION

HYPOTHESIS	Improving developer onboarding with a use-case-first selector will increase the first-successful-API-call rate.
CONTROL	Current experience — product-first docs homepage, developer self-diagnoses which API to use.
VARIANT	"What are you building?" selector routes directly to a recommended API stack with a pre-filled code sample.
PRIMARY	First successful API call rate within 24 hours of signup.
SECONDARY	7-day developer activation rate — at least one successful call on three or more distinct days.
GUARDRAIL	Support tickets asking "which API should I use," plus failed API call rate. If the selector routes people to the wrong stack, both rise.

14.2 Experiment 2 — Live credit / cost estimator

Does quantifying the free tier increase trial?

ACTIVATION

HYPOTHESIS	Showing a live cost estimate — "₹1,000 ≈ X requests" — at signup will increase signup-to-first-call conversion.
CONTROL	Current signup flow with static "₹1,000 free credits" messaging and no estimator.
VARIANT	Signup flow includes an interactive estimator tied to the developer's stated use case.
PRIMARY	Signup → first successful API call conversion rate.
SECONDARY	Time-to-first-call, measured as median minutes.
GUARDRAIL	Credit-exhaustion complaints and negative sentiment in Discord or support. An estimator that over-promises would show up here first.

14.3 Experiment 3 — Self-serve Growth plan

Will scaling developers buy a mid-tier?

REVENUE

HYPOTHESIS	Offering a self-serve "Growth" plan with defined SLAs will increase the upgrade rate among scaling developers who are not yet enterprise-ready.
CONTROL	Current model — free tier, then pay-as-you-go, then "contact sales" for anything beyond default rate limits.
VARIANT	A self-serve Growth tier with higher rate limits, priority support, and a defined lighter SLA, purchasable without a sales call.
PRIMARY	Percentage of developers exceeding default rate limits who upgrade to Growth within 30 days.
SECONDARY	Time from hitting the rate limit to resolution — either upgrade or churn.
GUARDRAIL	Enterprise-sales-qualified leads that self-select into Growth instead of a full enterprise contract. This is the cannibalisation risk, and it is the one that would kill the experiment.

14.4 Design principles applied

Every test has a guardrail

Not just a primary and secondary pair. Each guardrail protects against a specific unintended consequence — support load, misleading expectations, or revenue cannibalisation — and each is chosen because it would move *before* the damage became visible in revenue.

Every test moves the North Star

All three are designed to increase Weekly Successful AI Tasks, but through different levers: two through activation and one through revenue-tier expansion. If a proposed experiment could not name its lever, it did not make this list.

SEQUENCING NOTE

Experiments 1 and 2 can run concurrently on separate surfaces — the docs homepage and the signup flow. Experiment 3 should follow, because a mid-tier plan is only worth pricing once the activation improvements have widened the pool of developers who reach the rate limit at all.